

Centre for Quantum Control (QONTROL)

Research Tracks: RT1–RT4

Unlocking the potential of quantum systems

Department of Physics & Department of Chemistry
University of Oslo, Norway

2028–2037

- 1 Introduction: QONTROL at a Glance
- 2 RT1: Quantum Correlations
- 3 RT2: Quantum Dynamics
- 4 RT3: Quantum Readout
- 5 RT4: Method Development
- 6 Principal Investigators

QONTROL: Mission and Vision

Mission

Develop theoretical, computational, and experimental tools to **control quantum states** in atoms, molecules, and materials – underpinning the *second quantum revolution*.

Central scientific question

How can we engineer external forces to make quantum systems behave as we desire?

Key quantum concepts

Superposition • Interference • Entanglement

Systems covered

- Atoms & molecules
- Nanoscale systems / quantum dots
- Solid-state defects
- Quantum materials

Duration & host

2028–2037 (two 5-year periods)
University of Oslo (UiO)

Controllability

Drive any initial state to a chosen final state in finite time using admissible controls.

The Four Research Tracks

Track	Name	Core focus
RT1	Quantum Correlations	Many-body entanglement; strongly correlated systems
RT2	Quantum Dynamics	Time evolution; driven systems; decoherence mitigation
RT3	Quantum Readout	Measurement; sensing; metrology
RT4	Method Development	Algorithms, ML, open software, hardware benchmarking

Overall structure

All four tracks are **tightly coupled** and span three research areas: *Atoms & Molecules* • *Nanoscale Systems* • *Quantum Materials*.

Achieving quantum control requires: **(i)** preparing an initial state (RT1); **(ii)** steering its evolution (RT2); **(iii)** measuring the final state (RT3). RT4 provides the methodological backbone for all tracks.

RT1 – Quantum Correlations: Scope & Objectives

Scope

Addresses fundamental correlations in **many-body quantum systems**: atoms, molecules, quantum dots, and correlated materials.

Key scientific challenges

- Beyond mean-field toward **predictive many-body treatments**
- Entangled relativistic & non-relativistic electronic states
- Correlated electron–nuclear motion beyond Born–Oppenheimer
- Mechanisms of **defect–qubit** formation and exotic correlated-material phenomena

Beyond-state-of-the-art objectives

- 1 Realize and characterize superposition and entanglement in quantum systems
- 2 Explore non-classical correlations in chemical bond breaking and formation

Principal Investigators (Fig. 3)

Atoms & Molecules: T. B. Pedersen (TBP), M. Repišký (MR)

Nanoscale Systems: M. E. Bathen (MEB), S. Kvaal (SK)

Quantum Materials: J. Wells (JW), L. Vines (LV)

RT1 – Quantum Correlations: Research Highlights

Electrons on liquid helium and quantum dots(Hjorth-Jensen et al., PRX Quantum 2024)

Coulomb interactions tuned by electric fields to create **charge qubits with long coherence times**.
Exact experimental and theoretical treatment; basis for prototype quantum gates, circuits and sensors.

Beyond Born–Oppenheimer (Pedersen, Kvaal et al., JACS 2024)

First extraction of molecular and electronic structure from all-particle non-BO wave functions by stochastic sampling of conditional quantum probability distributions – essential for interpreting **attosecond quantum dynamics simulations**.

Silicon-vacancy defects in SiC (Bathen, Vines et al., npj QI 2019)

Theory, computation, and experiment to identify and control charge-transition levels and phonon couplings of **single-photon emitters and spin qubits** in 4H–SiC, advancing integration into quantum devices.

RT2 – Quantum Dynamics: Scope & Objectives

Scope

Understanding and **steering the time evolution** of quantum systems: free and driven dynamics, beyond few-level models, with relativistic effects.

Key scientific challenges

- Model **open quantum systems** with realistic environments
- Design **optimal inverse-control strategies**
- Mitigate **decoherence and noise**
- Time-dependent many-body theory for ionization processes
- ML-guided control from high-accuracy simulations

Beyond-state-of-the-art objectives

- 1 Theory and experiment for qubit time evolution in nanoscale systems
- 2 Reveal ultrafast quantum dynamics via attosecond probing and modeling

Principal Investigators (Fig. 3)

Atoms & Molecules: TBP, MR

Nanoscale Systems: MEB, SK

Quantum Materials: JW, LV

Attosecond chemistry

Laser pulses on the electron time scale enable control of chemical reactions via electron–nuclear entanglement.

Theoretical framework

Unified framework combining **theoretical modeling, simulation, and data-driven control**:

- Time-dependent many-body theory for open systems
- Optimal control theory (GRAPE, CRAB, ...)
- Lindblad / non-Markovian master equations
- Connects experimental observations with theory

Machine-learning guidance

- Reinforcement learning for optimal pulse design
- ML-guided control from high-accuracy simulations
- Programmable quantum evolution and robust qubits
- Generative models for control-landscape exploration

Key ambition

Develop time-dependent many-body theories for **open quantum systems including ionization processes**, creating a unified framework connecting experiment with theory.

RT3 – Quantum Readout: Scope & Objectives

Scope

Characterising quantum states through **measurement**: cornerstone of quantum computing, sensing, and metrology. Targets enhanced detection sensitivity via quantum correlations.

Key scientific challenges

- **Critical quantum sensing**: exploit proximity to quantum phase transitions for enhanced precision
- Study quantum material response to minute perturbations to optimise measurement fidelity
- Deepen the role of superposition, interference, and entanglement in quantum information processing
- *In operando* characterisation of prototype devices

Beyond-state-of-the-art objectives

- 1 Enhance sensitivity using superposition, interference, and entanglement
- 2 *In operando* characterisation of prototype quantum devices

Principal Investigators (Fig. 3)

Atoms & Molecules: TBP, MR

Nanoscale Systems: MEB, SK

Quantum Materials: JW, LV

Role across all tracks

RT3 specifically deepens the role of entanglement in **sensing and information processing**.

RT4 – Method Development: Scope & Objectives

Scope

Methodological and technological foundation for all other RTs: advances in experiment, computation, and data science.

Key activities

- Next-generation algorithms for static and time-dependent quantum-mechanical calculations
- **ML-enhanced modeling** of open quantum systems
- **Quantum-computing-based simulations**
- Open datasets and automated workflows
- State-of-the-art magnetic resonance and photoemission tools
- Generative deep learning and reinforcement learning

Beyond-state-of-the-art objectives

- 1 Synergetic development of new AI methods and many-body models for quantum control
- 2 Develop and benchmark experimental hardware for coherent control of nanoscale systems

Principal Investigators (Fig. 3)

Atoms & Molecules: TBP, MR

Nanoscale Systems: MEB, SK

Quantum Materials: JW, LV

Hylleraas Software Platform (HSP)

- Flexible **open-source software library** integrating quantum and classical dynamics codes with ML techniques
- High-performance computing framework
- All software and datasets **openly accessible** (gitlab.com/hylleraasplatform)
- Ensures reproducibility and interoperability

Machine learning integration

Pioneered ML-driven materials prediction and design, enabling rapid exploration of vast parameter spaces:

- Data-driven discovery and control
- Generative deep learning
- Agent based AI and Reinforcement learning for Quantum Control

Transformative potential

The systematic integration of quantum science and ML is **transformative**, enabling data-driven discovery across all research tracks and accelerating the second quantum revolution.

Principal Investigators

Name	Role	Dept	Key expertise
Morten Hjorth-Jensen	Director 2028–32	Physics	Quantum many-body theory; ML; quantum computing
Marianne E. BATHEN	Director 2033–37	Physics/ CMSE	Solid-state physics; expt quantum materials
Thomas Bondo Pedersen	Group Leader	Chemistry	Quantum many-body theory; laser-driven dynamics
Simen Kvaal	Group Leader	Chemistry	Quantum many-body theory; static & real-time theory
Michal Repišký	Group Leader	Chemistry	Relativistic electronic-structure; ReSpect code
Lasse Vines	Group Leader	Physics/CMSE	Semiconductor defect physics; expt quantum materials
Justin Wells	Group Leader	Physics/CMSE	Photoemission spectroscopy; expt quantum materials

All at University of Oslo (UiO). CMSE = Centre for Materials Science and Nanotechnology.

PI Research-Area Assignments (from Fig. 3)

Research Track	Atoms & Molecules	Nanoscale Systems	Quantum Materials
RT1 Quantum Correlations	TBP, MR	MEB, SK	JW, LV
RT2 Quantum Dynamics	TBP, MR	MEB, SK	JW, LV
RT3 Quantum Readout	TBP, MR	MEB, SK	JW, LV
RT4 Method Development	TBP, MR	MEB, SK	JW, LV
Centre Director (all RTs): MHJ = Morten Hjorth-Jensen			

Initials key

MHJ Morten Hjorth-Jensen **TBP** Thomas Bondo Pedersen **MR** Michal Repišký **MEB** Marianne E. Bathen
SK Simen Kvaal **JW** Justin Wells **LV** Lasse Vines

More people

Research Track	Atoms & Molecules	Nanoscale Systems	Quantum Materials
RT1 Quantum Correlations	ET, MC	SV, FM, CG, GL	FM, CG GL
RT2 Quantum Dynamics	ET	ET, FM, GL	FM, GL
RT3 Quantum Readout		FM, GL	FM, GL
RT4 Method Development	AN, DB, MC, ET, AC	SV, FM, GL, CG	FM, GL, CG

Initials key

SV Susanne Viefers	FM Francesco Massel	DB David Balcells	CG Cecilie Glittum	MEB Erik Tellgren
AC Antoine Camper	MC Michele Cascella	GL Gunnar Lange	AN Ainara Nova	